

Chapter-1

Introduction to Project Management

Definition of Project and Project Management

Project

- A project is an one time only set activities designed to attain specific objectives with the constraints of time, cost and quality through planned use of available resources so as to create a unique service.
- It is a planned set of interrelated tasks to be executed over a fixed period to provide some specific product as an output to the user.

Project Management

- Project management is the utilization of knowledge, skills, tools and techniques to power the planned project activities so as to meet the project requirements.
- The growth of project management focuses on goals, resources and schedule of each project.

Characteristics of a Project / Project Attributes

1. A project must specify its objectives clearly.
2. Each project has definite life cycle.
3. Projects are performed with the constraints of time, cost and quality.
4. Each project is unique.
5. A project operates in dynamic environment with flexibility to response changing environment.
6. All the projects must have some beneficiaries.



Triple Constraints of a Project

- The project is carried out under certain constraints - they are cost, time and scope.
- These three factors are called triple constraints.
- They can be explained as:
 1. Projects must be delivered within cost.
 2. Projects must be delivered on time.
 3. Projects must meet the agreed scope or objectives.
- The goal of the constraints is to meet customer quality requirements.

Inter relationships between the constraints

- If you reduce cost of project, you will either reduce its scope or increase its time.
- If you reduce time of project, you will either increase its cost or reduce its scope.
- If you increase scope of project, you will either increase its cost or time.

Project Objectives

- Project objectives is defined as the general statement of desired output or end result of the project.
- The objectives of any project must be: S M A R T
 1. Specific
 2. Measurable
 3. Attainable
 4. Realistic
 5. Time Bounded

Note : It is aggregated as *SMART*

Project Life Cycle

- Each project has its own life i.e. beginning and end.
- Generally life cycle of a project has four phases.
- The phases are:
 1. Project Formulation Phase
 2. Project Development Phase
 3. Implementation Phase
 4. Termination Phase

Project Formulation Phase

- This is the first phase of the project.
- It includes project identification and formulation.

Project Identification:

- In this stage, the project is developed.
- A project is born with creative ideas based on the environment, internal and external resources.

Project Evaluation:

- It involves formulation of the project to determine the constraints of the project.
- Statement of work and project proposal are developed.
- Scope of project, schedule, cost and quality of project are incorporated in statement of work.
- Project proposal is a initial document formed after conducting pre feasibility study and preliminary design of the project.

Project Development Phase

- It involves preparation of detailed plan with time, cost and quality.

Feasibility Study:

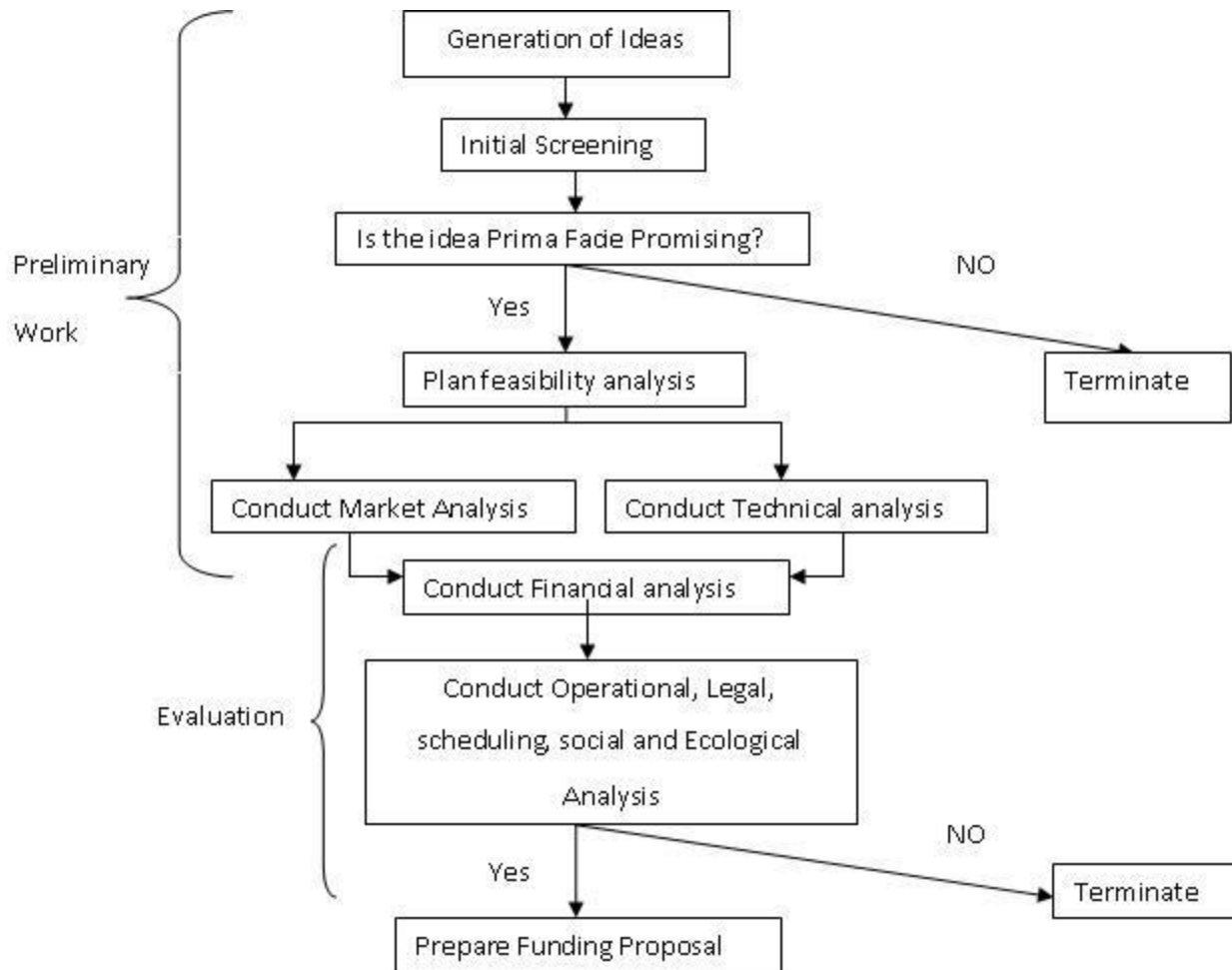
- It is the process of determining the implementability of the project.
- It involves technical, economic, market, management and environmental analysis.
- The schematic diagram of feasibility study is shown in given figure:

Appraisal:

- It is the evaluation of ability of project to succeed which is based on findings of feasibility study.
- It evaluates ability of project to achieve its objectives.
- It also involves comparison of the project with other projects.

Design:

- It involves preparation of project specification and implementation plan.



Implementation Phase

- It involves implementation and control of project activities.
- A project team is formed which is responsible to implement the project.
- Project manager is selected, who is responsible to mobilize project activities.
- Time to time evaluation and review of the project's progress is done.

Termination Phase

- The project is evaluated and finally handed over to the actual beneficiaries of the project.

Q) What are the key elements of project proposal?

1. Project objectives
2. Project activities
3. Execution process
4. Project schedule
5. Budget estimation
6. Evaluation and monitoring process

Chapter-2

Project Management Body of Knowledge

Understanding Project Environment

Project Environment

1. Internal Environment

- It is the elements present within the organization.
- It directly controls the actual project activities.
- It includes objectives, constraints, structure and resources.

2. Task environment

- It is the elements made up of stakeholders that surrounds the project.
- It only influences the project.
- It includes customer, contractor, supplier, competitor and so on.

3. External environment

- It includes political, legal, economic, socio-cultural and technological aspects of the environment.

PMBOK - Introduction

- PMBOK stands for Project Management Body of Knowledge.
- It is the set of standard terminology and guidelines for project management.
- It is the entire collection of processes, best practices and guidelines accepted as standards within the project management industry.
- It must be updated and disseminated over time.

Knowledge Contents Under PMBOK

1. Project Integration Management:

- Processes needed to identify, define, combine and coordinate various processes and activities.

2. Project Scope Management:

- Processes required to ensure that the project includes all the work required to complete the project successfully.

3. Project Time Management:

- Processes required to manage timely completion of the project.

4. Project Cost Management:

- Processes involved in planning, estimating, budgeting, funding, managing and controlling costs to complete project within approved budget.

5. Project Quality Management:

- Processes that determines project quality to ensure client's satisfaction.

6. Project Human Resource Management:

- Processes that organize, manage and lead the project team.

7. Project Communication Management:

- Processes required to ensure timely collection, creation, distribution, storage, retrieval, control, monitor and disposition of project information.

8. Project Risk Management:

- Processes to conduct risk analysis, identification, response planning and risk control on a project.

9. Project Procurement Management:

- Processes necessary to acquire products from outside the project team.

10. Project Stakeholder Management:

- Processes to identify people impacted by the project, analyze stakeholders expectations and engage stakeholders in project decision and execution.

General Management Skills: Effective and Ineffective Project Managers

- Project manager is the person who is incharge of overall planning and execution of a particular project.

Role of Project Manager

- They have responsibility of planning, procurement and execution of a project.
- They have to solve any issues that arise during the project life cycle.
- They have to analyze and view the project progress.
- They have to deal with the risks associated with the project and try to find optimal solutions to reduce them.
- They should ensure that all the team know about their roles.
- They should communicate with both team members and higher authority.
- They should manage project schedule and budget.

Effective Project Manager

The characteristics of an effective project manager are as follows:

1. Lead by example
2. Visionary
3. Technically competent
4. Decisive
5. Good Communicators
6. Good Motivators
7. Stands up to upper management when necessary
8. Supports team members
9. Encourages new ideas

Ineffective Project Manager

The characteristics of an ineffective project manager are as follows:

1. Set bad example.
2. Not self assured
3. Lacks technical expertise
4. Poor communicator
5. Poor motivator

Essential Interpersonal and Managerial Skills

Interpersonal Skills

- Communication
- Leadership
- Negotiation
- Adaptability
- Diligence

Managerial Skills

- Time Management
- Problem solving and critical thinking
- Budget estimation and usage
- Resource procurement
- Risk management
- Contract management

Q) What are the suggested skills for IT project managers?

- Technically sound (knowledge of platform, software and programs used by the organization)
- Problem solving using problem trees

Chapter-3

Portfolio and Project Management Institutes (PMI) Framework

Portfolio

- Project portfolio is the document that consists of key characteristics of the project.

Project Portfolio Management (PPM)

- It is a process used by project managers and project management organizations to analyze and manage current or proposed projects based on key characteristics.
- The objectives of PPM is to determine optimal resource delivery and to schedule activities to best achieve financial goals of an organization without degrading the constraints imposed by customers.

Benefits of PPM

- It helps to predict the outcome and plan for projects that will offer the best results.
- It helps to understand which risk of projects can offer the most rewards.
- The project manager gets the global view of each project which makes them able to predict the future problems.
- It provides a way to improve accuracy for estimating and managing financial resources of a group of projects.

Project Management Office

Project Management Institute (PMI) Framework

- PMI is a non profit organization which aims to bring standardization in the field of project management.
- The framework defines the principles of ethics and code of conduct for professionals.
- It focuses on continuous improvement, knowledge sharing and building network.
- It defines project management as a group of 10 knowledge areas, 5 process groups and 47 processes with well defined inputs, tools, techniques and output.

Reasons for failure of projects

- Improper requirement analysis
- Poor project management
- Lack of executive level support
- No linkage to business strategy
- Wrong team members
- No measures to evaluate project success
- No risk management
- Inability to adapt changes
- Lack of control and visibility

Drivers of Project Success

1. Strong leadership
2. Organizational maturity
3. Business-driven approach
4. Execution backing
5. User adoption
6. Visibility

Inhibitors of Project Success

1. Poor feasibility study
2. Poor requirement analysis
3. Lack of teamwork
4. Improper communication
5. No executive support
6. Visibility delay

Chapter-4

Project Management

Advantages of Project Management

1. Efficiency in delivering services
 - Good management provides better path to perform project activities.
 - It helps in timely completion of projects and yields greater productivity.
2. Improved customer satisfaction
 - Completing project on time and within budget make customer satisfied.
 - It acts as a base to build a strong customer network and increase organizational trust towards the customer.
3. Improved team development
 - Positive result from the project inspire the team members to perform better in future.
4. Competitive standing in market
 - Good project management helps to carry out project efficiently.
 - It helps in making good reputation in the competitive market.
5. Increased risk assessment
 - As all the members and their skills are properly analyzed, the possible risks can be detected and mitigated in time.

PM as per PMI

- Project management is the application of knowledge, skills and techniques to execute projects effectively and efficiently.
- It is a strategic competency for organizations, enabling them to tie project results to business goals.
- It defines 5 processes and 10 knowledge areas.

Project Management vs Project Portfolio Management

- Project management focuses on an individual project's path to completion whereas PPM considers every project or potential project and its viability to meet business goals.
- PPM provides the organization a view of upcoming, present and past projects which helps in analyzing to make sound decision.

Role and Responsibility of a Key Project Members

- Any project can not move in required direction without each of its key team members.
- The general roles of key team members are project manager, project team member, project sponsor, executive sponsor and business analyst.

Project Manager

- Develop a project plan
- Manage deliverables according to the plan
- Recruit project staff
- Lead and manage project team
- Determine methodology used on the project
- Establish a project schedule and determine each phase
- Assign task to team members

Project Team Member

- Contribute to overall project objectives
- Complete individual deliverables
- Provide expertise
- Work with users to establish business needs
- Document the process

Project Sponsor

- Approve project budget
- Ensure resource availability

Executive Sponsor

- Approve changes to project scope
- Provide additional funds for scope changes
- Approve project deliverables

Business Analyst

- Gather requirements
- Document technical and business requirements
- Verify that project deliverables meet the requirements

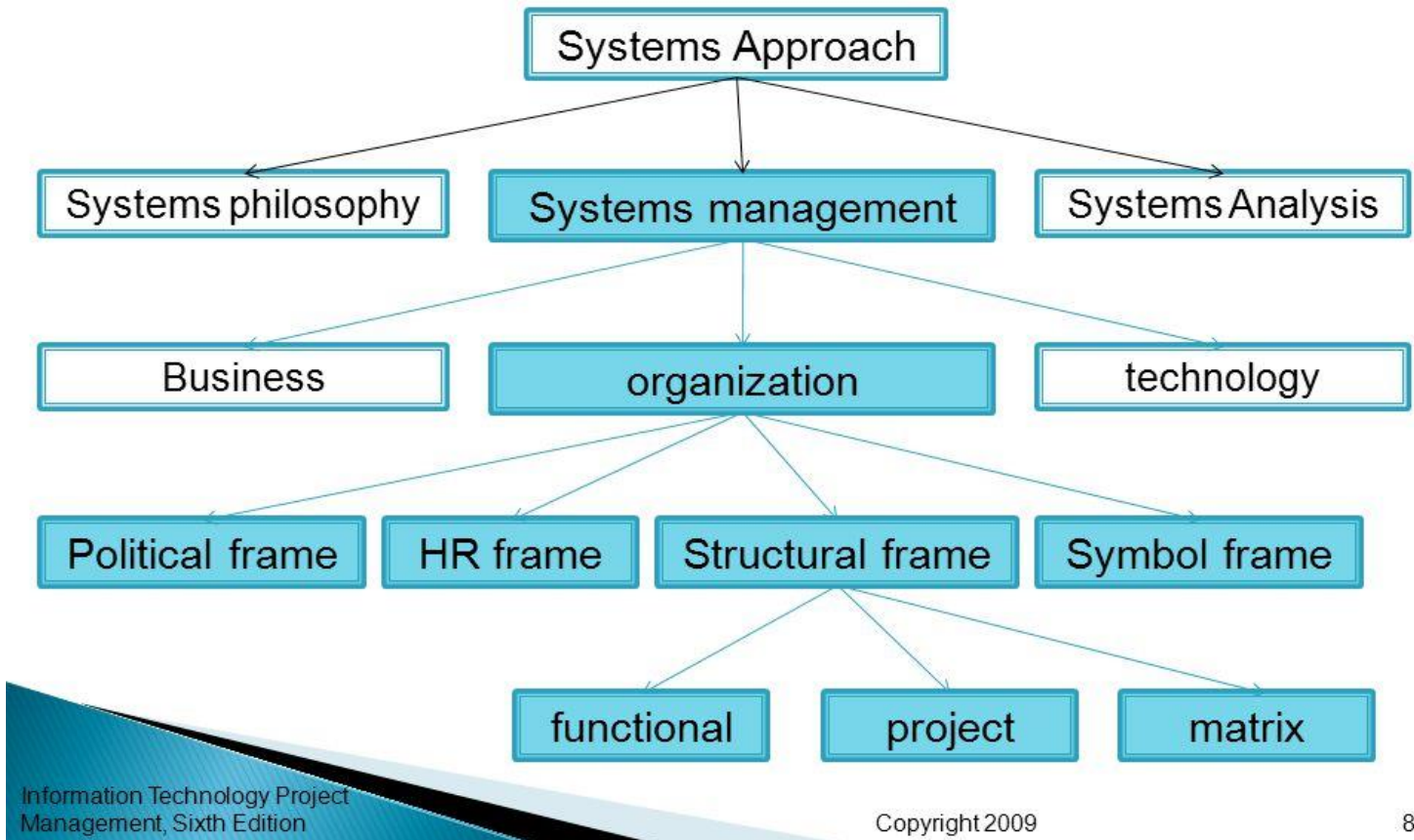
Chapter-5

Project and Organizational structure

System View of PM

- System approach is an analytical approach to project management.
- It includes three parts : system philosophy, system analysis and system management.
- System philosophy deals with overall model for thinking about things as systems.
- System analysis deals with problem solving which requires:
 1. defining scope of system
 2. dividing it into components
 3. identifying and evaluating problems
 4. examining alternative solutions
 5. identifying a satisfactory solution
- System management addresses business, technological and organizational issues before making changes to the system.

A systems approach

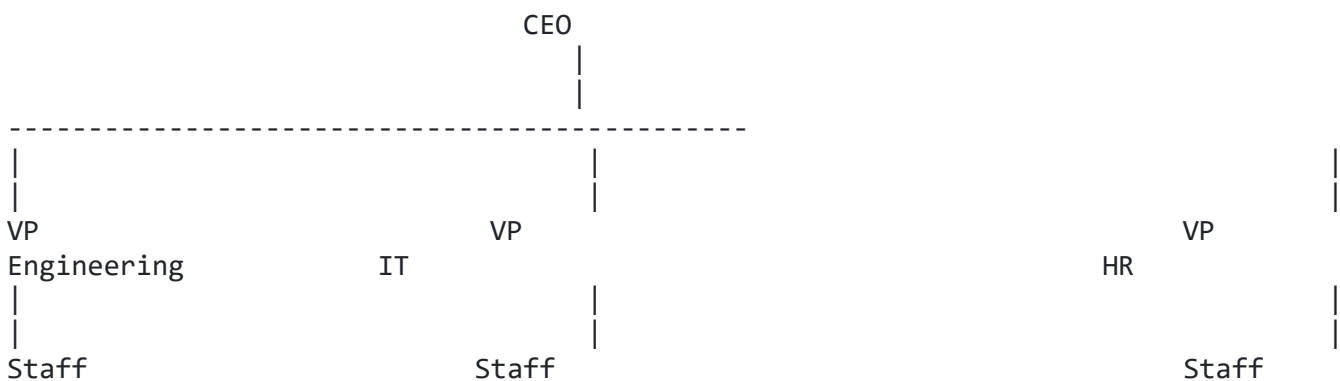


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Functional and Matrix Organization

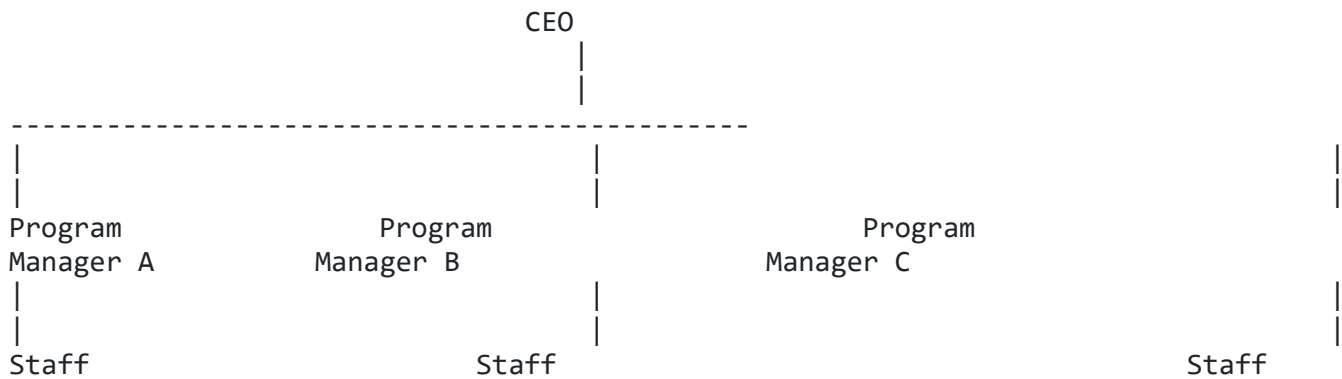
Functional Structure

- Functional managers report to the CEO.



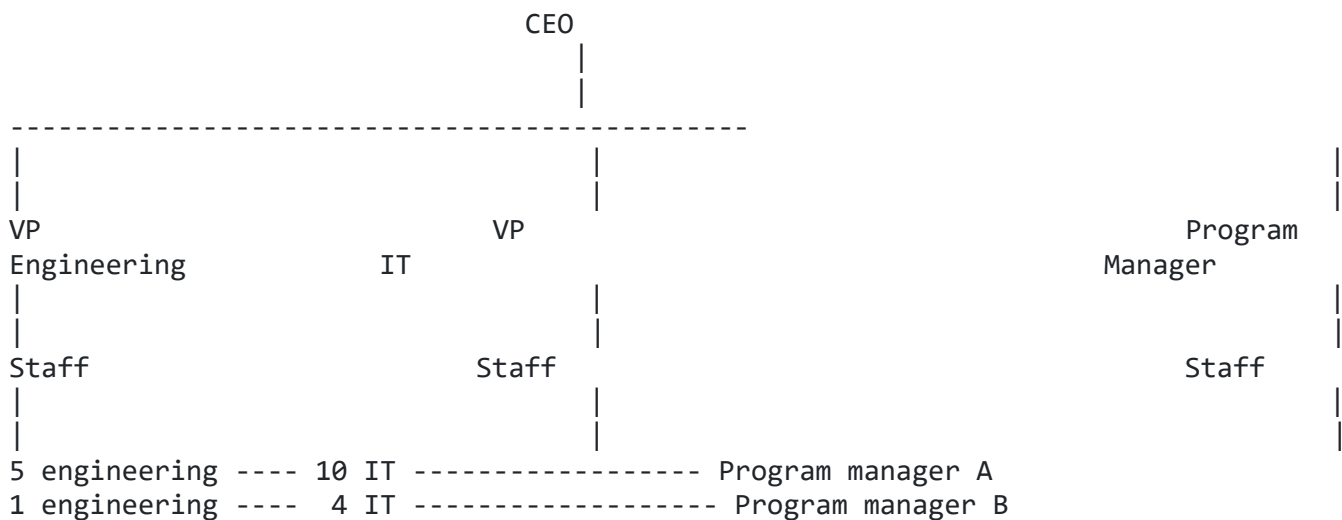
Project Structure

- Program managers report to the CEO



Matrix Structure

- It is middle ground between functional and project structure.
- A personnel reports to two or more bosses.



Organization Structure Influence on Projects

Project Characteristics	Functional	Weak Matrix	Balanced Matrix	Strong Matrix	Project
Project Manager's Authority	Little or More	Limited	Low to moderate	Moderate to high	High to almost total
Percentage of performing organization's personnel assigned full time to project work	Virtually none	0 - 25%	15 - 60%	50 - 95%	85 - 100%
Who controls project budget	Functional manager	Functional manager	Mixed	Project Manager	Project Manager
Project manager's role	Part time	Part time	Full time	Full time	Full time
Common title for project manager role	Project leader	Project leader	Project manager	Project manager	Project manager
PM administrative staff	Part time	Part time	Part time	Full time	Full time

Chapter-6

Project Management Process Groups

Project Management Processes

- A process is a series of actions that is performed to acquire some goals.
- Each process is provided with inputs which are analyzed using some standard tools and techniques so as to give desired outputs.
- Project management processes are concerned with describing and organizing the work of the project.
- Project management can be described by a number of inter-linked processes

Project Management Process Groups

1. Project Initiation:

- This process involves recognizing and starting a new project.
- The main focus is to select and start project
- The outputs of this process are:
 - a) Assigning project manager
 - b) Identifying key stakeholders
 - c) Completing a business case
 - d) Completing a project charter with signatures

2. Project Planning

- It consists of processes that perform establishment of scope, defining of objectives, developing project management plan and documenting the project activities.
- It is an iterative process group.
- The main focus is to prepare effective guide for project execution.
- The outputs of this process are as follows:
 - a) Team contract
 - b) Scope statement
 - c) Work breakdown structure
 - d) Project schedule
 - e) List of prioritized risks

3. Project Execution

- It consists of processes that is performed to complete the work specified in planning phase satisfying the user requirements.
- It involves proper team collaboration, resource allocation, integration of project activities and so on.
- It is also an iterative process.

4. Project Monitoring and Controlling

- It consists of processes that tracks and organizes the progress and performance of the project activities.
- Project performance is measured at regular intervals and analyze whether the project activities is directing towards the goals.
- It occurs during all phases of project life cycle.
- the outputs include performance reports, requested plan change and updates.

5. Project Closing

- It includes processes to conclude the completion of project.
- Customer acceptance determines this process group.
- The outputs are project archives and final report.

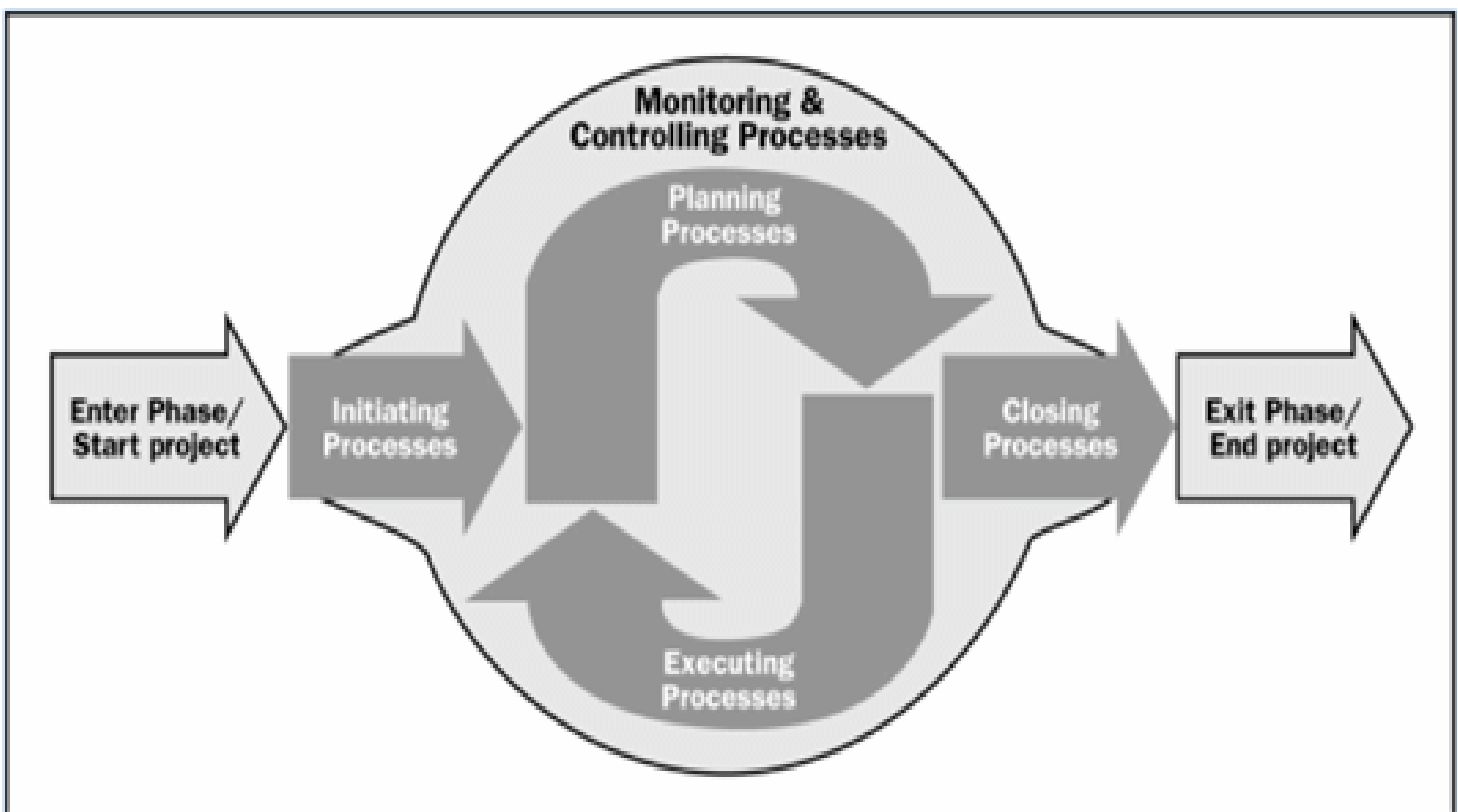


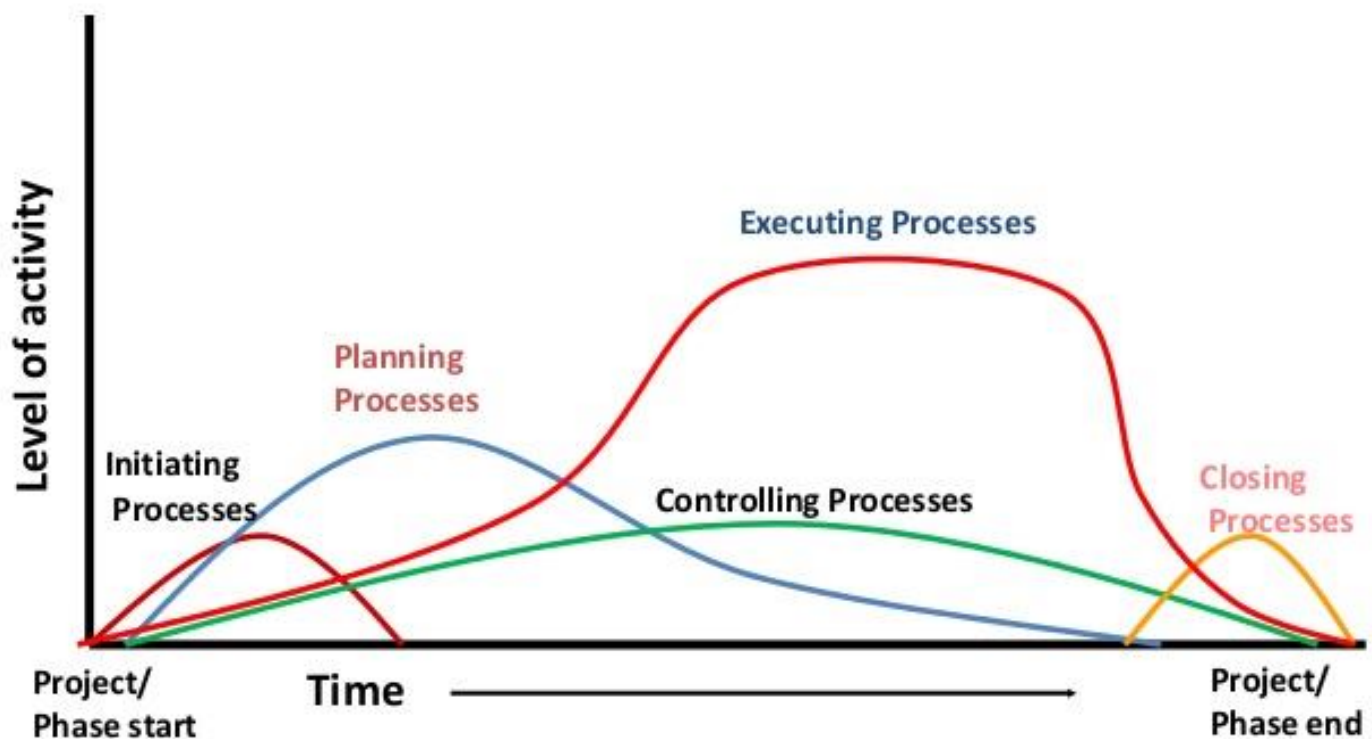
Figure 3-1. Project Management Process Groups

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Overlaps of Process Groups in a Phase

- It is possible that all process groups could be conducted within a phase of project life cycle.
- In each phase like concept development, feasibility study, prototype design and so on, all the process groups would be repeated.
- The overlaps is shown in given figure:

Overlap of process groups



Mapping of Project Management Process Group to Area of Knowledge

- Knowledge areas provide detailed description of the processes.
- The knowledge areas may cross various process groups.
- The mapping of process groups to knowledge areas are shown in given figure:

Knowledge Areas	Project Management Process Groups				
	Initiating Process Group	Planning Process Group	Executing Process Group	Monitoring and Controlling Process Group	Closing Process Group
4. Project Integration Management	4.1 Develop Project Charter	4.2 Develop Project Management Plan	4.3 Direct and Manage Project Work	4.4 Monitor and Control Project Work 4.5 Perform Integrated Change Control	4.6 Close Project or Phase
5. Project Scope Management		5.1 Plan Scope Management 5.2 Collect Requirements 5.3 Define Scope 5.4 Create WBS		5.5 Validate Scope 5.6 Control Scope	
6. Project Time Management		6.1 Plan Schedule Management 6.2 Define Activities 6.3 Sequence Activities 6.4 Estimate Activity Resources 6.5 Estimate Activity Durations 6.6 Develop Schedule		6.7 Control Schedule	
7. Project Cost Management		7.1 Plan Cost Management 7.2 Estimate Costs 7.3 Determine Budget		7.4 Control Costs	
8. Project Quality Management		8.1 Plan Quality Management	8.2 Perform Quality Assurance	8.3 Control Quality	
9. Project Human Resource Management		9.1 Plan Human Resource Management	9.2 Acquire Project Team 9.3 Develop Project Team 9.4 Manage Project Team		
10. Project Communications Management		10.1 Plan Communications Management	10.2 Manage Communications	10.3 Control Communications	
11. Project Risk Management		11.1 Plan Risk Management 11.2 Identify Risks 11.3 Perform Qualitative Risk Analysis 11.4 Perform Quantitative Risk Analysis 11.5 Plan Risk Responses		11.6 Control Risks	
12. Project Procurement Management		12.1 Plan Procurement Management	12.2 Conduct Procurements	12.3 Control Procurements	12.4 Close Procurements
13. Project Stakeholder Management	13.1 Identify Stakeholders	13.2 Plan Stakeholder Management	13.3 Manage Stakeholder Engagement	13.4 Control Stakeholder Engagement	

Table 3-1. Project Management Process Group and Knowledge Area Mapping

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Chapter-7

Project Integration Management

Project Charter

Integration Management

- It is the knowledge area that consists of different processes to ensure that all different project components are properly coordinated.
- Integration is necessary to:
 1. manage change and communication
 2. reduce project time and cost
 3. involve stakeholders
 4. make results visible
 5. identify problems and solutions easily.

Integration Processes

1. Develop Project Charter
2. Develop project management plan
3. Direct and manage project execution
4. Monitor and control project work
5. Perform integrated change control
6. Close project or phase

Project Charter

- Project charter is a statement of objectives of a project that sets out detailed project goals, roles and responsibilities, identify stakeholders and level of authority.
- It is a document with respect to proposal.

Roles:

- It documents reasons for undertaking the project.
- It outlines objectives and constraints of the project.
- It provides solutions to the problem.

Benefits:

- It improves customer relationships.
- It acts as a tool that improves project management process.
- It improves communication.
- It helps to gain project sponsorship.

Elements of Project Charter

Input	Tools and techniques	Output
• Project statement of work • Enterprise environmental factors • Contract • Organizational process assets	• Project selection method • Project management methodology • Project MIS • Expert Judgement	• Project charter

Project Management Plan; Direct and Manage Project Execution; Monitor and Control Project Work

Project Management Plan

Input	Tools and techniques	Output
• Preliminary project scope statement • Project management process • Enterprise environmental factors • Organizational process asset	• Project management methodology • Project MIS • Expert Judgement	• Project Management Plan

Direct and Manage Execution

Input	Tools and techniques	Output
• Project Management Plan • Approved corrective action • Approved preventive action • Approved change request • Approved defect repair • Validated defect repair • Administrative closure procedure	• Project management methodology • Project MIS	• Deliverables • Requested changes • Implemented change control • Implemented corrective action • Implemented preventive action • Implemented defect repair • Work performance information

Monitor and Control Project Work

Input	Tools and techniques	Output
• Project management plan • Work performance information • Rejected change request	• Project management methodology • Project MIS • Earned value management • Expert judgement	• Recommended corrective action • Recommended preventive action • Forecast • Recommended defect repair • Requested changes

Integrated Change Control

- It is the process of reviewing all change requests; approving and managing changes to deliverables, assets, documents and project management plan; and communicating their disposition.
- It focuses on measuring the impact of change on all project limitations.
- Requested changes are accepted or rejected based on evaluation and impact.

Input	Tools and techniques	Output
• Project Management Plan • Requested changes • Work performance information • Recommended preventive action • Recommended corrective action • Recommended defect repair • Deliverables	• Project management methodology • Project MIS • Expert judgement	• Approved change request • Rejected change request • Project management plan update • Project scope statement update • Approved corrective action • Approved preventive action • Approved defect repair • Validated defect repair • Deliverables

Close Project; Project Scope Management

Close Project

Input	Tools and techniques	Output
• Project management plan • Contract documentation • Enterprise environmental factors • Organizational process asset • Work performance information • Deliverables	• Project management methodology • Project MIS • Expert judgement	• Administrative closure procedure • Contract closure procedure • Final Product • Organizational process asset update

Scope Management

- It is the knowledge area that includes processes required to ensure that the project includes only the work required.
- Without proper scope of a project, time schedule and budget can not be managed.
- It includes following processes:
 1. Collect requirements
 2. Define scope
 3. Create work breakdown structure
 4. Verify scope
 5. Control scope

Collect Requirements

- Requirements are the needs to be fulfilled from the project.
- It should be included considering other factors like quality, compliance and so on.
- It involves detailed study of project requirements.
- If any miss occurs in this process, it may cause project failure.

Input	Tools and techniques	Output
• Project charter • Stakeholder register	• Interview • Focus group • Facilitated workshop • Group creativity • Group decision making • Questionnaire and survey • Observation • Prototype	• Requirement document • Requirement management plan • Requirement traceability matrix

Define Scope

Input	Tools and techniques	Output
• Project charter • Requirement documentation • Organizational process asset	• Expert judgement • Product analysis • Alternative identification • Facilitated workshop	• Project scope statement • Project document update

- Statement of work is a detailed narrative description of work required for a project.
- It contains introduction and background, technical description, timeline and milestones and client expectations.

Work BreakDown Structure

Create WBS

- Work breakdown structure (WBS) is a process to set a project's scope by breaking down its overall mission into a consistent set of specific tasks.

Input	Tools and techniques	Output
• Project scope statement • Requirement documentation • Organizational process asset	• Decomposition	• WBS • WBS Dictionary • Scope baseline • Project document update

Importance of WBS

- It echoes project objectives.
- It offers logical structure.
- It establishes a method of control.
- It improves communication.

Decomposition

- Decomposition is the breaking down of the deliverables into more manageable parts.
- The work is decomposed into work packages.
- A work package is the lowest level of WBS which is output oriented.

Example:

New Warehouse

```
|
|
-----
||
||
Design - 1 Construction - 2
||
|-----Structural - 1.1 |
||-----> Steel package 1.1.1 |-----> Concrete install - 2.1
||-----> Concrete package 1.1.2 |
||
|-----> Architectural - 1.2 |-----> Steel install - 2.2
|
|-----> Elevations 1.2.1
```

Scope Verification and Control

Verify Scope

Input	Tools and techniques	Output
• Project management plan • Requirement documentation • Requirement traceability matrix • Validated deliverables	• Inspection	• Accepted deliverables • Change request • Project document update

Control Scope

Input	Tools and techniques	Output
• Project management plan • Work performance information • Requirement documentation • Requirement traceability matrix • Organizational process asset	• Variance analysis	• Work performance measure • Organizational process asset • Change request • Project management plan update • Project document update

Chapter-8

Project Cost Management

Cost and Project

Project Cost Management

- It is the knowledge area that consists of processes involved in planning, estimating, budgeting, financing, funding, managing and controlling costs so as to complete the project within budget.

Cost Estimation; Types of Cost Estimate

Estimate Cost

Input	Tools and techniques	Output
• Enterprise environmental factor • Organization process asset • Project scope statement • WBS • WBS dictionary • Project management plan	• Analogous estimating • Determine resource cost rate • Bottom up estimating • Parametric estimating • Project management software • Vendor bid analysis • Reserve analysis • Cost of quality	• activity cost estimate • Activity cost estimate supporting detail • Requested change • Cost management plan update

Importance of cost estimation

- It provides standard that can be used to compare and control the expenditures during the course of the project.
- It helps the project manager to allocate scarce resources as the project progresses.

Classification of Project Cost

1. Direct vs Indirect Cost

- Direct cost is the cost that can be charged against the project. Eg: cost of manpower and materials used.
- Indirect cost is the cost indirectly involved in the project. Eg: overhead cost, selling and administrative cost.

2. Recurring and Non-recurring Cost

- Recurring cost is the cost that are repeatedly incurred throughout the project lie cycle.
- Non recurring cost is the one time cost that are incurred at the beginning or at the end of the project.

3. Fixed vs Variable Cost

- Fixed cost is the cost that do not vary with usage. Eg: Cost incurred in purchase of capital equipment
- Variable cost is the cost that vary directly with the usage. Eg: labor cost.

4. Normal vs Expedited Cost

- Normal cost is the cost incurred when project tasks are completed according to the original planned duration.
- Expedited cost is the crash cost which are unplanned incurred as a result of steps taken to accelerate project completion.

Cost Estimating Methods

1. Ballpark estimate

- It is done when there is no sufficient information or time available.
- It helps in rough estimate of resources needed for a project in initial phase.
- It should attempt to have 30% accuracy.

2. Feasibility estimate

- It is done after preliminary work is completed.
- It is generally performed when information on material cost is available.
- It should attempt to have 10% accuracy.

3. Definitive estimate

- It is done after completion of design work.
- It requires clear understanding of scope and capabilities of the project.
- It attempts to have 5% error margin.

4. Comparative estimate

- It uses historical data of previous project as a reference for estimating cost of current project.

5. Parametric estimating

- It uses mathematical model for estimation.

6. Vendor bid analysis

- It estimates using bids and allowances for gaps in bid slope.

7. Reserve analysis

- It adds contingency to each activity cost estimate as zero duration item.

Cost Budgeting; Cost Aggregation; Deriving Budget from Activity Cost

Determine Budget

- Budgeting is the process of planning for the cost of project resources.
- Budget implies the level of management support.
- Budget is estimated as:

Materials + Labor + Equipment + Capital + Overhead + Profit = Bid

Input	Tools and techniques	Output
• Project scope statement • WBS • WBS dictionary • Activity cost estimate • Project schedule • Resource calendar • Contract • Cost management plan	• Cost aggregation • Reserve analysis • Parametric estimation • Funding limit reconciliation	• Cost baseline • Project funding requirements • Cost management plan updates • Requested changes

Types of Budgeting

1. Top - down

- Top managers estimate overall budget for the project.
- The estimates are broken down at each lower level.

2. Bottom - up

- Low level managers estimate budget for each work package.
- Overhead and profits are added to develop project budget.

3. Negotiated

- It is the combination of top down and bottom up budgeting.
- Both are prepared and compared.
- Any differences are negotiated.

4. Activity based costing

- Assign cost to activities that uses resources.
- Identify cost drivers associated with activity.
- Compute cost rate per cost deliver unit.
- Multiply cost drive rate times the volume of cost driver units used by the project.

Budget Contingency

- It is the allocation of extra funds to cover uncertainties.
- It provides protection against unknown and uncertain elements.

Cost Control Process; Cost Control Methods

Control Costs

Input	Tools and techniques	Output
• Cost baseline • Project funding requirements • Performance report • Work performance information • Approved change requests • Project management plan	• Cost change control system • Performance measurement analysis • Forecasting • Project performance reviews • Project management software • Variance management	• Cost estimate update • Cost baseline update • Performance measurement • Forecasted completion • Requested changes • Recommended corrective actions • Organizational process asset updates • Project management plan update

EV Management and Benefits; Variance Analysis

Earned Value Management (EVM)

- EVM is a project management technique for measuring project performance and progress in an objective manner.
- It helps to find variances in projects based on comparison of work performed and work planned.
- It is used on cost and schedule control.

Equations

Assumptions:

1. Planned value (PV) = budgeted cost
2. Earned value (EV) = actual work completed
3. Actual cost (AC) = cost incurred
4. Estimate to complete (ETC) = what is left
5. Estimate at completion (EAC) = what final cost will be

Analysis:

1. Cost variance (CV) = $EV - AC$
 - Negative = under budget
 - Positive over budget
2. Schedule variance (SV) = $EV - PV$
 - Negative = behind schedule
 - Positive = ahead of schedule
3. Cost Performance Index (CPI) = EV / AC
 - How much we get for every dollar we spend?
4. Schedule Performance Index (SPI) = EV / PV
 - Progress as % against plan
5. ETC = $EAC - AC$
 - How much more we have to spend?
6. Variance at completion (VAC) = $BAC - EAC$
7. EAC = BAC / CPI

Q) If EV is twice its AC of the project, calculate CPI and CV. Is the project over or under budget?

Let AC = x, then:

$$EV = 2x$$

Now,

$$CPI = EV / AC = 2x / x = 2$$

Also,

$$CV(\%) = (EV - AC) / AC * 100\%$$

$$= (2x - x) / x * 100\%$$

$$= 100\%$$

As, EV-AC is positive, the project is under budget.

Q) If SPI is 0.75 with earned value being 60. Calculate planned value and also state whether the project is ahead or behind schedule?

Given:

$$SPI = 0.75$$

$$EV = 60$$

We know that,

$$SPI = EV / PV$$

$$\text{or, } PV = 60 / 0.75$$

$$\text{or, } PV = 80$$

Now,

$$\text{Schedule variance (SV)} = EV - PV = 60 - 80 = -20$$

As, SV is negative, the project is behind schedule.

Q) A project is scheduled for time of 12 months with estimated cost of 400000. After 3 months, evaluation is done and identified that 40% of work is accomplished with cost of 200000. Calculate cost and schedule variance for the project.

Given:

Planned schedule = 12 months

Estimated cost = 400000

Planned value = Estimated cost = 400000

At the end of 3 months:

$PV = 400000/12 * 3 = 100000$

Project completed = 40%

$EV = 40\% \text{ of } 400000 = 160000$

$AC = \text{Cost incurred} = 200000$

Now,

$\text{Cost variance (CV)} = EV - AC = 160000 - 200000 = - 40000$

As, CV is negative, the project is over budget.

$\text{Schedule variance (SV)} = EV - PV = 160000 - 100000 = 60000$

As, SV is positive, the project is ahead of schedule.

Chapter-9

Project Quality Management

Quality Theories

- Software quality management ensures that the appropriate level of quality is achieved by submitting improvements to the product development process.
- The activities related to quality management are as follows:
 1. Quality assurance
 2. Quality planning
 3. Quality control

Quality Planning; Project Quality Requirements; Cost of Quality

- Quality planning is the process of identifying which quality standards are relevant to the project and determining how to satisfy them.
- It involves planning how to fulfill product quality requirements.
- It has following principles:
 1. Quality is determined by requirements of customers.
 2. It is better to avoid mistakes than to repair the defects.
 3. Cost of quality must be approved by management.

Input	Tools and techniques	Output
• Project scope statement • Project management plan • Enterprise environmental factors • Organizational process asset	• Cost benefit analysis • Benchmarking • Design of experiment • Cost of quality	• Quality management plan • Quality metrics • Quality checklists • Improved project management plan

Project Quality Requirements

- It defines the customer's expectations for quality, internal attributes that indicates quality factors being satisfied or not and measures to be used to give visibility to the levels of quality being achieved.
- It is defined in terms of quality factors, quality criteria and quality metrics.
- It provides an exact and measurable definition of quality for the specific project.

Cost of Quality

- Cost of quality is the total cost incurred by investment in preventing non conformance to quality requirements.

Quality Management Plan

The quality management plan should address following attributes:

- Responsibility of management
- Document management
- Requirement scope
- Design control
- Testing and quality assurance
- Quality audits
- Training requirements

Quality Assurance; Quality Audit

Quality Assurance

- It is a process that is used to ensure the quality of products or services.
- A constant effort is made to enhance quality practices in the organization, so that continuous improvements are expected.
- A dedicated quality assurance team is formed.
- The QA team has following responsibilities:
 1. Define a process and procedure to achieve and improve quality.
 2. The team should go through formal industrial training.
 3. Ensure traceability of test cases to requirements.

Quality Audit Process

- Audit is a systematic, independent and documented process of obtaining audit evidence and evaluating objectively to determine the extent to which audit criteria are fulfilled.
- An internal audit is an audit performed by the members of an organization.
- Audit is performed because of following reasons:
 1. To ensure compliance with ISO.
 2. To ensure compliance with organization requirements.
 3. To ensure QMS is effectively implemented.

Input	Tools and techniques	Output
• Policy • History • Documents • Customer feedback • Planning • Information	• Comparative analysis • Checklist	• Audit report • Follow up • Improvement • Conclusion

Quality Control Process; Test Life Cycle

Quality Control

- It is the process of monitoring specific project results to determine whether it comply to relevant quality standards and identifying ways to eliminate causes of unsatisfactory performance.

Input	Tools and techniques	Output
• Quality management plan • Quality metrics • Quality checklist • Work performance information	• Control chart • Flow chart • Histogram • Pareto chart • Statistical sampling • Defect repair review	• Quality control measurement • Validated defect repair • Recommended corrective action • Recommended preventive action • Recommended defect repair • Updated project management plan

Control Chart

- It is a statistical process control tool used to determine if a manufacturing or business process is in a state of control.
- If analysis of control chart indicates the process is currently under control, no corrections to control parameters are needed.

Pareto Chart

- It is a chart containing both bar graph and line graph.
- Individual values are represented in descending order by bars.
- Cumulative total is represented by line.
- It is used to highlight the common sources of defects, highest occurring type of defect and so on.

Total Quality Management (TQM)

- It is an approach for project quality management that takes everything related to quality under consideration.
- In TQM, the processes that produce services are thoroughly managed minimizing process variation and providing predictable quality level.
- The principles used in TQM are as follows:
 1. Top management
 2. Training needs
 3. Customer orientation
 4. Involvement of employees
 5. Corporate culture
 6. Tools and techniques
 7. Continuous improvements
- The tools and techniques used for TQM are as follows:
 - Pareto chart
 - Scatter plot
 - Control chart
 - Fishbone and Ishikawa diagram
 - Histogram
 - Check list
 - Check sheets
 - Assessment of customer need
 - Market analysis
 - Competition analysis
 - Work flow analysis
 - Logistic analysis

Software Testing Life Cycle

1. Requirement Analysis
2. Test Planning
3. Test Case Development
4. Environment Setup
5. Test Execution
6. Test Cycle Closure

Q) Write possible steps to improve quality of IT project.

- Define quality to match user needs.
- Broadcast simple quality metrics.
- Fine tune goals to include quality.
- Get the requirements right.
- Test smarter to test less.
- Design applications to reduce bug risks.
- Optimize the use of testing tools

Chapter-10

Project Communication Management

Importance of Communication Management

- A project is a team work rather than individual assessment. So, for a successful completion of any project, there must be proper coordination between team members, project managers and top level managers.
- Communication management is the knowledge area that employ the processes to ensure timely and appropriate generation, collection, distribution, storage, retrieval and disposition of project information.
- The importance of communication management are as follows:
 1. It focuses on keeping communication relevant to the project.
 2. It defines the mediums for participants to communicate.
 3. It ensures all the people get the right information at right time.
 4. It tracks the way for remote communication.
 5. It keeps records of all the communication details for future purposes.
 6. It ensures all the participants establish accountability towards the information they perceived.

Communication Planning; Communication Requirement Analysis

Communication Planning

- It is the process of determining the information and communication needs of the project stakeholders.

Input	Tools and techniques	Output
• Project scope statement • Project management plan • Enterprise environmental factors • Organizational process assets	• Communication requirement analysis • Communication technology	• Communication management plan

Communication Requirements

- It defines which participants should require which information to move towards project success.

Communication Management Plan

- It is a document describing:
 1. the stakeholder's communication requirements
 2. which information has to be communicated with whom
 3. frequency of communication
 4. ways of communication
 5. guidelines for meetings.

Conducting Effective Meeting

The guidelines for conducting effective meetings are as follows:

1. Identify purpose and desired outcomes of the meeting.
2. Identify and invite only the appropriate people.
3. Ensure presence of participants and their contributions.
4. Develop proper agenda to be discussed.
5. Establish the ground rules for meeting.
6. Proper time management in meeting.
7. Designate a note taker to prepare minutes.
8. Evaluation of the meeting process.
9. Follow up by providing readable minutes to all attendees.

Information Distribution Process

- It is the process of making information available to project stakeholders in a timely manner.
- It includes communication management plan and responds to unexpected information requests.

Input	Tools and techniques	Output
• Communication management plan	• Communication skills • Information gathering and retrieval system • Information distribution system	• Updated organizational process assets • Requested changes

Communication Skills

- It ensures that right person gets right information at right time.
- It emphasizes on sender to make information clear and complete so that the receiver can obtain the correct information.

Performance Reporting; Integrated Reporting System

Performance Reporting

- It is the process of collecting and distributing performance information like status, progress and so on.
- The report may be forecast report, progress report, status report or variance report.

Input	Tools and techniques	Output
• Work performance information • Performance measurements • Forecasted completion • Quality control measurement • Project management plan	• Information presentation tools • Status review meetings • Time reporting system • Cost reporting system	• Performance report • Forecast • Recommended corrective action

Integrated Reporting System

- Integrated report is a concise communication about how an organizational strategy, governance and performance lead to the creation of value over time.
- It provides greater context for performance data and helps in decision making to be long term.
- It helps to complete financial and sustainability reports.

Q) Differentiate horizontal and vertical communication.

- Horizontal communication occurs between people of same hierarchy level. Vertical communication occurs between people of different hierarchy level.
- Horizontal communication aims to coordinate activities of various divisions. Vertical communication aims to provide necessary directives and feedback.
- Horizontal communication enjoys greater degree of freedom. Vertical communication must be formal.

Chapter-11

Project Risk Management

Risk and Project Risk

Risk

- Risk is an uncertainty which is associated with a future and may or may not occur and a corresponding potential for loss.

Types of Risk in Software Projects

1. Schedule related planning risk

- It involves risks related to running behind schedule.

- The causes are as follows:

- a) incorrect time estimation
- b) improper resource allocation and utilization
- c) unexpected scope expansion

2. Financial risk

- It is the monetary risk associated with budget overruns.

- The causes are as follows:

- a) improper budget estimation
- b) scope expansion
- c) improper finance tracking

3. Operational risk

- It is the risk associated with operational activities of the project.

- The causes are as follows:

- a) improper process implementation
- b) conflict in team
- c) lack of clarity in responsibility
- d) communication breakdown
- e) lack of training

4. Functional risk

- It is the technical risk related to the functionality of the software.

Risk Management

- It is the knowledge area that include process concerned with conducting risk management planning, identification, analysis, response and control on a project.
- It follows pattern of plan, execute and control.

Steps in risk management

1. Plan the risk management
2. Identify the risks
3. Analyze risk qualitatively
4. Analyze risk quantitatively
5. Develop risk response options and actions
6. Monitor whether risks become facts

Risk Management Planning Process; Risk Management Plan

- It is the process of deciding how to plan and execute risk management activities for a project.

Input	Tools and techniques	Output
• Project scope statement • Project management plan • Enterprise environmental factors • Organizational process assets	• Planning meetings and analysis	• Risk management plans

Risk Identification Techniques

- It is the process of determining the risks that may hamper the project and documenting their effects.
- It is an iterative process as risk can not be foreseen at once.

Input	Tools and techniques	Output
• Project scope statement • Project management plan • Risk management plan • Enterprise environmental factors • Organizational process asset	• Documentation reviews • Information gathering technique • Checklist analysis • Assumption analysis • Diagramming technique	• Risk register

Qualitative Risk Analysis Process

- It is the process for prioritizing risks for further analysis by assessing and combining their probability of occurrence and impact.

Input	Tools and techniques	Output
• Project scope statement • Risk management plan • Risk register • Organizational process asset	• Risk probability and impact assessment • Probability and impact matrix • Risk categorization • Risk urgency assessment	• Updated risk register

Quantitative Risk Analysis Process

- It is the process of numerically analyzing the effect on overall project objectives of identified risk.
- It is performed on prioritized risks obtained after qualitative risk analysis.

Input	Tools and techniques	Output
• Project scope statement • Project management plan • Risk management plan • Risk register • Organizational process asset	• Sensitivity analysis • EVM analysis • Decision tree analysis • Simulation and modeling	• Updated risk register

Sensitivity Analysis / Tornado Diagram

- It helps to view which variables have most impact on a project objective.
- It is a bar chart showing comparative sensitivity analysis output.
- The longer the bar, more sensitive the project objective is to the risk.
- Risk are presented in descending order.

Decision Tree Analysis

- It helps to visualize uncertainty and various decision option.
- It is a tree that shows comparative analysis of various decisions for project risks.
- It helps in determining the best alternatives to minimize impacts of risks.

Risk Response Planning

- It is the process of developing options and actions to enhance opportunities and to reduce impacts of risk on project goals.

Input	Tools and techniques	Output
• Risk management plan • Risk register	• Strategies for negative risks (Avoidance, Transfer risk effects to third party, Mitigation) • Strategies for positive risks (Exploit, Share, Enhance)	• Updated risk register • Updated project management plan • Risk related contractual agreements

Risk Monitoring and Control Process

- It is the process of tracking identified risks, monitoring residual risks, identifying new risks and executing risk response plans.

Input	Tools and techniques	Output
• Risk management plan • Risk register • Work performance information • Performance reports	• Risk reassessment • Risk audit • Variance and trend analysis • Reserve analysis • Status meeting	• Updated risk register • Recommended corrective actions • Recommended preventive actions • Updated project management plan

Chapter-12

Project Procurement Management

Procurement Management Process Flow

- Procurement management is the knowledge area that deals with the process to purchase the products or services from outside the project team to perform the work.
- It includes internal structure following the pattern of plan, execute and control.

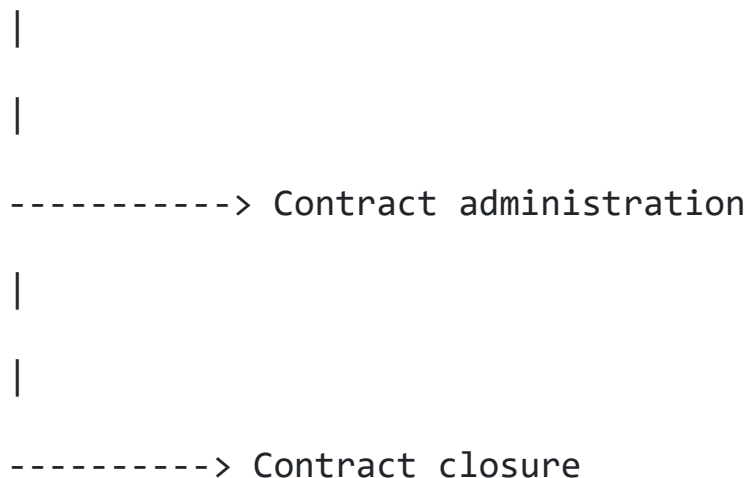
Steps in procurement management

1. Determine what to purchase; when and how
2. Documents service requirements and identify potential sellers.
3. Requests for seller response.
4. Selects a seller
5. Administrates the contract
6. Organizes the contract closure

Plan purchase



Select seller



Plan Purchase and Acquisition Process

- It is the process of determining what to purchase when and how.
- The proper project needs should be identified and procured properly.

Input	Tools and techniques	Output
• Project scope statement • WBS • WBS Dictionary • Project management plan • Enterprise environmental factors • Organizational process asset	• Make or buy analysis • Expert judgement • Time and material contract • Contract types	• Procurement management plan • Contract statement of work • Make or buy decision • Requested changes

Contract Types

1. Fixed price contract:

- Purchase order for a specified item to be delivered by a specified date for a specified time.
- This can not manage scope changes.

2. Cost reimbursable contract:

- Buyer pays all real costs of the seller plus a fee.
- It can manage scope changes.

Plan Contracting Process

- It is the process of documenting service requirements and identifying potential sellers.
- It aims to generate procurement documents and establish evaluation criteria.

Input	Tools and techniques	Output
• Procurement management plan • Contract statement of work • Make or buy decisions • Project management plans	• Standard forms • Expert judgement	• Procurement documents • Evaluation criteria • Updated contract statement of work

Standard Forms : Tools

- Standard contracts
- Standard descriptions of procurement items
- Non-disclosure agreements
- Proposal evaluation criteria checklists

Procurement Documents

- It is a document which describes target and process of selecting a seller.
- It is used to seek proposals from prospective sellers.

Evaluation Criteria

- It is the set of conditions which must be fulfilled by the bidder for being able to be selected.
- It focuses on how well can the seller able to fulfill the statement of work and how low is the price the seller wants to have.

Request Seller Response Process

- It is the process of obtaining information, quotations, bids or proposals from the sellers on how project requirements can be met.

Input	Tools and techniques	Output
• Procurement management plan • Procurement documents	• Bidder conference • Advertising • Develop qualified seller list	• Qualified seller list • Procurement document package • Proposals

Select Seller Process

- It is the process of reviewing offers, choosing among potential sellers and negotiating written contract with each other.

Input	Tools and techniques	Output
• Procurement management plan • Evaluation criteria • Procurement document package • Proposals • Qualified seller lists • Risk register	• Weighting system • Independent estimate • Screening system • Contract negotiation • Seller rating system • Expert judgement • Proposal evaluation technique	• Selected seller • Contract • Contract management plan • Resource availability • Updated procurement management plans

Contract Administration Process

- It is the process to manage contract and relationship between buyer and seller, reviewing and documenting how a seller is performing.
- It ensures buyers and sellers both perform as per the contract.

Input	Tools and techniques	Output
• Contract • Contract management plan • Selected seller • Performance report • Work performance information	• Contract change control system • Buyer conducted performance review • Inspection and audit • Payment system • Claim administration • Record management system	• Contract documentation • Requested changes • Recommended corrective actions • Updated procurement management plan • Updated contract management plan

Contract Closure Process

- It is the process responsible to complete and settle each contract.
- It also verifies that all work and deliverables are acceptable.

Input	Tools and techniques	Output
• Procurement management plan • Contract management plan • Contract documentation • Contract closure procedure	• Procurement audits • Record management system	• Closed contract • Updated organizational process asset

Chapter-13

Developing Custom Processes for IT Projects

Developing IT Project Management Methodology

1. Access current and optimal level of project management maturity. It helps to get current status and future possibility to project management maturity. It provides benchmark against future improvements.
2. Gather and document all existing templates and processes.
3. Get project team members together and maps out the process for how projects should work.
4. Fill the blanks of what is missing.
5. Build your methodology iteratively by using your team to document process flow chart.
6. Use your methodology as planned.
7. Conduct audits to see if methodology is being used as expected.

Moving Forward with Customized Management Process

Necessity of Custom process for IT Projects

1. Role of customer is dominant.
 2. Environment is more dynamic.
 3. Tasks are unique and complex.
 4. Role of government is less clear.
 5. Sources of knowledge are different.
-

Outsourcing and Offshoring

- Outsourcing occurs when a company contracts a specific process out to a third party, finding someone who specializes in whatever needs to be done.
- Offshoring occurs when businesses send in-house jobs overseas.
- Offshoring sends jobs out of the country.
- Both saves organizational finance.

Six Sigma

- It is a set of tools and techniques for process improvement.
- It seeks to improve quality of the output of a process by identifying and removing the causes of defects and minimizing variability.
- It uses empirical and statistical quality management methods.
- It follows patterns Plan-Do-Check-Act cycle.
- The methodologies used are as follows:
 1. DMAIC (Define, Measure, Analyze, Improve, Control)
 2. DMADV (Define, Measure, Analyze, Design, Verify)

Certified Associate in Project Management

- CAPM is a credential offered by PMI.
- It is an entry level certification for project practitioners.
- It is intended to demonstrate candidate's understanding of fundamental knowledge, terminology and processes of effective project management.

Project Management Maturity

- It is a formal tool used to measure an organizational project management maturity.
- It provides a path to take advancement of PM maturity once the initial level of maturity and areas for improvement are identified.
- It provides logical path for progressive development.
- It follows software engineering institute's Capability Maturity Model's five evolutionary maturity levels and examines maturity development across ten knowledge areas.

Maturity levels

Level 1 - Initial Process

Level 2 - Structured process and Standards

Level 3 - Organizational standards and Institutionalized process

Level 4 - Managed process

Level 5 - Optimizing process

Future Trend

1. Greater emphasis on leadership and communication skills.
2. Talent retention
3. Agile framework
4. Increasing complexity of IT projects
5. Remote project works

Chapter-14

Balanced Scorecard and ICT Project Management

Balanced Scorecard

- Balanced scorecard is a strategy performance management tool.
- It is a semi standard structured report used by managers to keep track of the execution of activities by the staff within their control and to monitor the consequences arising from these actions.
- The critical characteristics of the balanced scorecard are as follows:
 1. It focuses on strategic agenda of the organization.
 2. It selects small number of data items to monitor.
 3. It uses both financial and non-financial data items.
- It also helps in improving communication and feedback process.
- The areas of consideration of the balanced scorecard are as follows:
 1. Financial perspective
 2. Business process perspective
 3. Customer perspective
 4. Learning and growth perspective
- It creates a right balance between the components of organizational objectives and vision.

Need of Balanced Scorecard

1. Increases focus on business strategy and its outcomes.
 2. Leads to improvised organizational performance.
 3. Align workforce to meet organizational strategy.
 4. Targets key drivers of future performance.
 5. Improves level of communication.
-

Chapter-15

FORMS OF OWNERSHIP

The different types of business ownership are: -

1. Single Ownership (Private Undertaking).
2. Partnership.
3. Joint Stock Company
4. Cooperative Organization (Or Societies)
5. Public Sector
6. Private Sector.

1. Single Ownership:

Ownership when applied to an industrial enterprise means title to and possession of the assets of the enterprise, the power to determine the policies of operation, and the right to receive and dispose of the proceeds.

It is called a single ownership when an individual exercises and enjoys these rights in his own interest. A business owned by one man is called single ownership. Single ownership does well for those enterprises which require little capital and lend themselves readily to control by one person.

Examples of enterprises run by single owner are printing press, auto repair shop, wood working plant, a small fabrication shop, etc., Le., retail trades, service industries and small engineering firms. In single ownership, one person contributes the original assets to start the business, maintains and controls business operations, reaps full benefit in terms of profit and is fully liable for all debts associated with the business.

Advantages of Single Ownership:

1. Easy to establish as it does not require to complete any legal formality.
2. Simplicity of organization.
3. The expenses in starting the business are minimal.
4. Owner is free to make all decisions.
5. This type of ownership is simple, easy to operate and extremely flexible.

6. The owner enjoys all the profits, thus,
7. There is a great deal of personal motivation and incentive to succeed.
8. Minimum legal restrictions are associated with this form of ownership.
9. Owner can keep secrecy as regards the raw materials used, method of manufacture, etc.
10. Single ownership associates with it the great ease with which the business can be discontinued.

Disadvantages of Single Ownership:

1. The owner is liable for all obligations and debts of the business.
2. The business may not be successful if the owner has limited money, lacks ability and necessary experience to run the business.
3. Because of relatively unstable nature of the business, it is difficult to raise capital for expanding the business.
4. If the business fails, creditors can take the personal property as well as the business property of the (single) owner to settle their claims. This means single ownership involves unlimited liability for debts and losses.
5. There is limited opportunity for employees as regards monetary rewards (e.g., profit sharing, bonuses, etc.) and promotions.
6. Generally, single ownership firm has limited life, i.e., the firm may cease to exist with the death of the proprietor. This is the cause of unstable nature of the firm (refer 3 above).

Applications of Single Ownership:

Single ownership is suitable:

1. For retail trades, service concerns and small engineering firms which require relatively small capital to start with and to run.
2. For those businesses which do not involve high risks of failure.
3. When the business can be taken care by one person.

2. Partnership:

A single owner becomes inadequate as the size of the business enterprise grows. He may not be in a position to do away with all the duties and responsibilities of the grown business. At this stage, the individual owner may wish to associate with him more persons who have either capital to invest, or possess special skill and knowledge to make the existing business still more profitable.

Such a combination of individual traders is called Partnership. Partnership may be defined as the relation between persons who have agreed to share the profits of a business carried on by all or any of them acting for all.

Individuals with common purposes join as partners and they put together their property, ability, skill, knowledge, etc., for the purpose of making profits. In brief, partnership is an association of two or more (up to 20) persons to carry on as co-owners of a business for profit.

Partnerships are based upon a partnership agreement which is generally reduced to writing. It should cover all areas of disagreement among the partners. It should define the authority, rights and duties of each partner. It should specify- how profits and losses will be divided among the partners, etc.

Kinds of Partners:

- (i) Active Partners who take active part in the management of the business enterprise.
- (ii) Sleeping Partners who do not take any active part in the conduct of the business. Both Active and Sleeping partners are responsible for the debts of the Partnership.

General Duties of Partners:

Partners should:

- (i) Be just and faithful to one another.
- (ii) Render true accounts and full information about everything that affects any partner.
- (iii) Cooperate and accommodate each other.
- (iv) Have confidence in each other and better mutual understanding.
- (v) Respect the views of one-another.

Types of Partnership:

(i) General Partnership:

Whatever has been discussed above so far pertains to General Partnership; besides that in a general partnership, each partner has full agency powers and may bind the partnership by any act, i.e., each partner may act as though he were an individual proprietor.

General partnership differs from single ownership in that the actions of any partner not only affect himself but they affect other partners also. As the partnership grows or personnel changes occur, additional partners can be had with the consent of all old partners.

Advantages of General Partnership:

- (i) Large capital is available to the firm.
- (a) The firm possesses much better talents, judgment and skills.
- (iii) General partnership is easy to form and is relatively inexpensive in terms of organization cost.
- (iv) Incentive for success is high.
- (v) There is a definite legal status of the firm.
- (vi) Partners have full control of the business and possess full rights to all profits.
- (vii) Partnership associates tax advantages with it.
- (viii) Partnership firms can borrow money quite easily from the banks.
- (ix) For all losses, there are more than one person to share them.

Applications of General Partnership:

General Partnership does very well in

Law firms,

Retail trade organization,

Medical clinics,

Small engineering firms, etc.

Disadvantages of General Partnership:

- (i) Each partner has unlimited liability for the debts of the firm,
- (ii) Danger of disagreement and distrust among the partners,

- (iii) Authority being divided among the partners,
- (iv) Partnership lacks permanence and stability; it has limited life. Partnership may dissolve if a partner dies,
- (v) Investors and lenders hesitate to provide money because of the lack of stability of a partnership firm, and
- (vi) All partners suffer because of the wrong steps taken by one partner.

(ii) Limited Partnership:

Limited partnership type of ownership overcomes the two main disadvantages [e.g. number (i) and (v) mentioned above] of general partnership. Limited partnership is an association of one or more general partners who manage the business and one or more limited partners whose liability is limited to the capital they have invested in the business.

Limited partners share the profit but they do not participate or interfere with the control or management of the firm. Moreover, limited partners have their liabilities limited to the amount of their investment.

Thus, those investors and lenders who used to hesitate investing in the venture can do so without much risk. Limited partnership type of ownership is easy and less costly to form, and personal incentive to succeed is retained. A disadvantage associated with limited partnership is that the limited partner, though he invests in the business, has no voice in the management.

3. Joint Stock Company:

Joint Stock Company overcomes many of the disadvantages associated with Partnership types of industrial ownership, such as:

- (i) Difficulties in raising capital,
- (ii) Easy disruption,
- (iii) Lack of facility for centralized management, and
- (iv) Unlimited liability, etc.

A joint stock company is an Association of individuals, called shareholders, who join together for profit and agree to supply capital divided into shares that are transferable for carrying on a specific business. Death, insolvency, disablement or lunacy of the shareholders does not affect the joint stock company. A joint stock company consists of more than twenty persons for carrying any business other than the banking business.

These persons give a name to the company, mention the purpose for which it is formed, and state the nature and the amount of capital (shares) to be issued, etc., and submit the proposal to the Registrar of Companies. As the registrar issues a certificate in this connection, the company starts operating. The managing body of a joint stock company is Board of Directors elected by the shareholders.

The Board of Directors:

- (i) Makes policies;
- (ii) Takes decisions; and
- (iii) Runs the company efficiently.

The liability of the members (or shareholders) of a joint stock company is limited to that capital only of which they hold the shares. Finance is raised by issuing shares, debentures, bank loans, loans from industrial and finance corporations.

Types of Joint Stock Company:

There are two types of joint stock companies:

(a) Private Limited Company:

- (i) The capital is collected from the private partners; some of them may be active while others being sleeping.
- (ii) Private limited company restricts the right to transfer shares, avoids public to take up shares or debentures.
- (iii) The number of members is between 2 and 50, excluding employee and ex-employee shareholders.
- (iv) The company need not file documents such as consent of directors, list of directors, etc., with the Registrar of Joint Stock Companies.
- (v) The company need not obtain from the Registrar, a certificate of commencement of business.
- (vi) The company need not circulate the Balance Sheet, Profit and Loss Account, etc., among its members; but it should hold its annual general meeting and place such financial statements in the meeting.
- (vii) A private company must get its accounts audited.
- (viii) A private company has to send a certificate along with the annual return to the Registrar of Joint Stock Companies stating that it does not have shareholders more than fifty excluding the employee and ex-employee shareholders.

Actually, a private joint stock company resembles much with partnership and has the advantage that big capital can be collected, then could be done so in partnership.

(b) Public Limited Company:

- (i) In Public limited company, the capital is collected from the public by issuing shares having small face value (Rs. 50,20,10).
- (ii) The number of shareholders should not be less than seven, but there is no limit to their maximum number.
- (iii) A public limited company has to file with the Registrar of Joint Stock Companies, documents such as consent of the directors, list of directors, director's contract, etc., along with the memorandum of association and articles of association.
- (iv) A public company has to issue a prospectus to the public.
- (v) It has to allot shares within 180 days from the date of prospectus.
- (vi) It can start only after receiving the certificate to commence business.
- (vii) It has to hold a Statutory Meeting and to issue a Statutory Report to all members and also to the Registrar within a certain period.
- (viii) There is no restriction on the transfer of shares. (be) Directors of the company are subject to rotation.
- (x) The public company must get its account audited every year by registered auditors.
- (xi) It has to send financial statements to all members and to the Registrar.
- (xii) It has to hold a general meeting every year.
- (xiii) The Managing Agent gets a fixed percentage of net profit as remuneration.

Advantages of Joint Stock Companies:

- (i) A huge sum of money can be raised.
- (ii) It associates limited liability with it.
- (iii) Shares are transferable.
- (iv) Company's life is not affected by the life (death) of shareholders.
- (v) Services of specialists can be obtained.
- (vi) Risk of loss is divided among many shareholders.
- (vii) The company associates with it stability, efficiency and flexibility of management.

Disadvantages of Joint Stock Companies:

- (i) A good deal of legal formalities is required for the formation of a joint stock company.
- (ii) Company is managed by big shareholders only.
- (iii) High paid officials manage the whole shows; they cannot have as high interests in the company as the proprietors can have.
- (iv) People can commit frauds with the company.
- (v) Board of directors and managers who remain familiar with the financial position of the company may sell or purchase shares for their personal profits.
- (vi) It is difficult to maintain secrecy as in partnership.
- (vii) The team spirit with which partnership works, is lacking in a joint stock company.
- (viii) Divided responsibility.

Applications of Joint Stock Companies:

- (i) Steel mills,
- (ii) Fertilizer factories, and
- (iii) Engineering concerns, etc.

4. Cooperative Organization (Or Societies):

It is a form of private ownership which contains features of large partnership as well as some features of the corporation. The main aim of the cooperative is to eliminate profit and provide goods and services to the members of the cooperative at cost.

Members pay fees or buy shares of the cooperative, and profits are periodically redistributed to them. Since each member has only one vote (unlike in joint stock companies), this avoids the concentration of control in a few hands.

In a cooperative, there are shareholders, a board of directors and the elected officers similar to the corporation. There are periodic meetings of shareholders, also. Special laws deal with the formation and taxation of cooperatives.

Cooperative organization is a kind of voluntary, democratic ownership formed by some motivated individuals for obtaining necessities of everyday life at rates less than those of the market. The principle behind the cooperative is that of cooperation and self-help.

Forms of Cooperative Enterprises:

- (i) Consumer's Cooperatives, in retail trade and services.
- (ii) Producer Cooperatives, for group buying and selling such items as dairy products, grain, fruit, etc.
- (iii) Cooperative farming for more and good quality yield from the farms.
- (iv) Cooperative housing for constructing and providing houses to the members of the association at relatively lesser rates.
- (v) Cooperative credit society, to provide loans to the needy individuals.

Advantages of Cooperative Enterprises:

- (i) Daily necessities of life can be made available at lower rates.
- (ii) It is the democratic form of ownership.
- (iii) Overheads are reduced as members of the cooperative may render honorary services.
- (iv) It promotes cooperation, mutual assistance and the idea of self-help.
- (v) The chances of large stock-holding (hoarding) and black marketing are eliminated.
- (vi) No one person can make huge profits.
- (vii) Common man is benefited by cooperatives.
- (viii) Monetary help can be secured from government.
- (ix) Goods required can be purchased directly from the manufacturers and therefore can be sold at less rates.

Disadvantages of Cooperative Enterprises:

- (i) Since the members of the cooperative manage the whole show, they may not be competent enough to make it a good success.
- (ii) Finance being limited; specialist's services cannot be taken.
- (iii) Conflict may arise among the members on the issue of sharing responsibility and enjoying authorities.
- (iv) Members who are in position may try to take personal advantages.
- (v) Members being in services may not be able to devote necessary attention and adequate time for supervising the works of the cooperative enterprise.

5. Public Sector:

Concept of Public Sector:

A public enterprise is one that is:

- (1) Owned by the state,
- (2) Managed by the state, or
- (3) Owned and managed by the state.

The sector of public enterprises is popularly known as the Public Sector. Public enterprises are controlled and operated by the Government either solely or in association with private enterprises. Public enterprises are controlled and operated by the Government to produce and supply goods and services required by the society. Ultimate control of public enterprises remains with the state and the state runs it with a service motto.

Its sphere embraces all units, irrespective of risks involved and profit expected. There is no dearth of capital in public sector and business expansion is not difficult. Public sector prevents concentration and unbalanced growth of industries.

Public sectors are accountable in terms of their results to Parliament and State Legislature. A public enterprise is seldom as efficient as a private enterprise; wastage and inefficiency can seldom be reduced to a minimum.

Evolution of Public Sector:

The Industrial Revolution gave rise to many bitter social evils. It also gave birth to private capitalism. Consumers and workers were exploited and, therefore, there arose the need of State Intervention in industrial field. The intervention led to evolution of public sector/enterprises. The evolution of public sector in India is recent.

Prior to 1947, there was virtually no public sector barring the field of transport and communication, i.e., Railways, Posts and Telegraphs etc., are being managed by the Central Government since pre-independence period. Since independence, a large number of public enterprises have been established both by Central and State Governments.

The Hindustan Shipyard, The Hindustan Steels, the Hindustan Machine Tools, The Bharat Heavy Electricals, Indian Telephone Industries, Indian Airlines, Life Insurance Corporation are a few examples of public sector.

Objectives of Public Sector:

- (1) To provide basic infrastructure facilities for the growth of economy.
- (2) To promote rapid economic development.
- (3) To undertake economic activity strategically important for the growth of the country, which if left to private initiative would distort the national objective.
- (4) To have balanced regional development and even dispersal of economic activity throughout the country.
- (5) To avoid concentration of economic power in a few hands.
- (6) To create employment opportunities on an increasing scale.
- (7) To earn foreign exchange in order to export commodities not available in the country e.g., petroleum oil, sophisticated weapon systems etc.
- (8) To look after well-being and welfare of public.
- (9) To minimize exploitation of workers and consumers.

Merits of Public Sector:

- (1) Public sector helps in the growth of those industries which require huge amount of capital and which cannot flourish under the private sector.
- (2) Public sector helps in the implementation of the economic plans and enables them to reach the target of achievement within a prescribed period by taking initiative in- the establishment of industries of its own accord.
- (3) Due to the absence of profit motive in the public sector, the consumers are benefitted by greater, better and cheaper products.
- (4) Public enterprise prevents the concentration of wealth in the hands of a few and paves the way for equitable distribution of wealth among different sections of community.
- (5) Public enterprise encourages industrial growth of under-developed regions in the country.
- (6) Profits earned by public sector may be used for the general welfare of the community.
- (7) Public sector offers equitable employment opportunities to all; there is no discrimination, as may be in a private sector.
- (8) Capital, raw material, fuel, power and transport are easily made available to them.

Demerits of Public Sector:

- (1) Public sector can rarely attain the efficiency of a private enterprise; wastage and inefficiency can seldom be reduced to a minimum.
- (2) Due to heavy administrative expenses, state enterprises are mostly run at a loss leading to additional burden of taxation on the people.
- (3) There is too much interference by the Government and Politicians in the internal affairs of the public enterprises. As a result, inefficiency increases.
- (4) Delay in decisions is a very common phenomenon in public enterprises.
- (5) Incompetent persons may occupy high levels.
- (6) Workers (unlike in private concerns) shirk work.

6. Private Sector:

Private sector serves personal interests and is a non-government sector. Profit (rather than service) is the main objective. Private sector constitutes mainly consumer's goods industries where profit possibilities are high. Private sector does not undertake risky ventures or those having low-profit margin. Private enterprises are run by businessmen; capital is collected from the private partners.

Merits of Private Sector:

- (i) The magnitude of profits incurred is high.
- (ii) The efficiency of the private enterprise is high,
- (iii) Wastage of material and labor is minimum.
- (iv) Decision-making is very prompt.
- (v) There is no interference in its internal affairs by politicians or Government.
- (vi) Competent persons occupy high levels.

Demerits of Private Sector:

- (i) There is exploitation motive, the workers and the consumers may not receive fair deal.
- (ii) There is dearth of capital to expand the business.
- (iii) Private enterprise leads to concentration of wealth in the hands of a few.
- (iv) Private enterprises lead to unbalanced growth of industries.